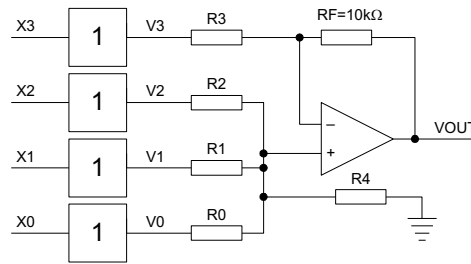


Problem Sheet 5 – Data Converters (Lectures 10 and 12)

(Question ratings: A=Easy, ..., E=Hard. All students should do questions rated A, B or C as a minimum)

- 1B. A $3\frac{1}{2}$ digit Digital Voltmeter has a display range of ± 1999 and an accuracy of ± 2 on the display. How many bits would a binary A/D converter need to have for its ± 0.5 LSB accuracy to be as good as that of the DVM?
- 2B. A 12-bit converter has a resolution of 1 mV (i.e. 1 LSB = 1 mV) and input voltages in the range ± 0.5 mV are converted to the value 0. What range of input voltages will be converted to -2047 ?
- 3B. A 10-bit converter converts an input voltage x to the value $\text{floor}(x / 10\text{mV})$. If $1\text{ V} < x < 8\text{ V}$, what range of output values will be obtained?



- 4C. $X3:0$ is a 4-bit signed number whose value, X , lies in the range -8 to $+7$. If the logic levels of $V3:0$ are 0 V and +5 V, choose values for $R0$ to $R4$ so that V_{OUT} is equal to $X/8$ volts.
- 5C. The composite video signal to drive a monochrome TV monitor takes one of three different voltages according to the values of two digital signals DATA and SYNC:

DATA	SYNC	V_{OUT}
0	0	0.0
1	0	0.7
0	1	-0.3
1	1	Don't Care

Design a circuit to generate V_{OUT} having a $50\ \Omega$ output impedance. You may assume that output logic levels are 0 and 5 V and that +5 V and -5 V power supplies are available should you need them. You do not need any op-amps although you will need at least one logic gate.

- 6B. Signals on a compact disc are stored as sequences of 16-bit numbers. Determine the maximum undistorted signal-to-noise ratio obtainable for a music signal whose peak amplitude is 10 times as great as its RMS value.
- 7C. A sample-and-hold circuit is used to store the input voltage of a 12-bit A/D converter during each conversion. The sample-and-hold circuit has an aperture uncertainty of 5 ns and a leakage current of ± 1 nA. The A/D converter has an input voltage range of ± 10 V.
 If the input voltage is a sine wave of amplitude 10 V, calculate the input frequency at which the aperture uncertainty will result in an error of ± 0.5 LSB [surprisingly low].
 If the sample-and-hold uses a storage capacitor of 200 pF calculate how long the input voltage can be held before it changes by 0.5 LSB due to the leakage current.